

# Hemanth Kumar

## Machine Learning Engineer

[hpeddameti1s@semo.edu](mailto:hpeddameti1s@semo.edu)  
<https://www.linkedin.com/in/1-hemanth-kumar>  
<https://github.com/heManKuMAR6>  
+1-959-999-5443  
Saint Louis, Missouri, USA

Experienced Data Science & AI Engineer with a 7-year track record in deep learning, federated learning, reinforcement learning, and MLOps. Skilled in integrating research with scalable cloud-based model deployment on AWS, Azure, GCP, and algorithm design with Industry standards.

## Work Experience

### Graduate Assistant & Machine Learning Researcher

Jan 2024 - Present

*Southeast Missouri State University*

- Designed federated learning pipelines using Flower and TensorFlow to enable diagnostic model training across hospital nodes, achieved 93% global accuracy while ensuring zero raw data sharing.
- Contributed to machine learning projects by writing and optimizing coding scripts for various algorithms and Applied statistics to analyze large datasets and extract meaningful insights for machine learning models.
- Orchestrated distributed training for autonomous vehicle perception models using Horovod and MultiWorkerMirrored- Strategy, reducing training time by 40% on cloud-based GPU clusters.
- Authored and have done Technical Reviews for two IEEE research papers on privacy-preserving federated inference and communication-efficient architectures in healthcare AI systems.

### AI/ML Engineer

Jan 2022 - Aug 2024

*Ford Motor Company*

- Engineered object detection models using YOLOv5 with domain-adaptive training; boosted pedestrian/cyclist detection accuracy to 91% mAP under adverse weather conditions and reduced false negatives by 30%.
- Built an embedded vision pipeline using DeepLabV3+ and MobileNetV3 for real-time lane and road sign recognition; achieved 50 FPS with latency under 30ms on NVIDIA Jetson Xavier.
- Developed a reinforcement learning-based supply chain optimization engine using Ray RLlib and SimPy; reduced delivery delays by 15% and inventory costs by 12% in simulated logistics.
- Managed code repositories and tracked changes using GIT, Kubernetes, and Helm, employing version control best practices for developing machine learning models for surveillance data analysis.
- Built XGBoost-powered predictive maintenance models using telemetry data from 10K+ vehicles; integrated with Kafka and Airflow to enable real-time anomaly alerts with 98.7% precision.

### Data Scientist

Jan 2021 - Dec 2022

*Deloitte*

- Automated financial reporting and anomaly detection on SAP/Oracle financials using Airflow and Isolation Forests, reduced reporting time from 4 hours to 15 minutes with high audit readiness.
- Architected scalable fraud detection pipelines handling 3M+ transactions daily using PySpark, CatBoost, and SHAP explainability; achieved 93% precision and 95% recall in identifying anomalous behavior.
- Built predictive time-series models (LSTM, ARIMA) leveraging Bloomberg data; enabled real-time volatility forecasting that accelerated decision-making by 40% for client portfolios.
- Devised learning objectives for machine learning projects, facilitated research and development, and helped streamline change management processes for smooth model deployment and user adoption.
- Applied intuitive problem-solving techniques to enhance AI/ML model performance and accuracy and assisted with the translation of complex datasets for AI/ML model training and evaluation.

**AI/ML Consultant**  
*Enque Private Solutions*

Jan 2020 - Dec 2021

- Built reinforcement learning models using PPO in RLlib for dynamic smart grid energy switching; improved energy efficiency by 13% across simulated city environments.
- Developed robotic vision modules using ROS, YOLOv7, and SLAM for warehouse automation; increased item retrieval speed by 25% and reduced human errors in inventory tracking by 40%.
- Assisted in the Curation of datasets for machine learning model training and evaluation and contributed to UX research and testing, gathering insights to improve AI/ML product usability.

**Computer Vision Engineer**  
*Cognizant*

Jan 2018 - Dec 2020

- Leveraged transfer learning to adapt YOLOv8 and EfficientDet models, efforts reduced training time by 40% and improved detection of rare defect types within two weeks.
- Designed and deployed TensorFlow-based video analytics pipeline for lip sync and artifact detection using FFmpeg; enhanced content QA by 60% and achieved 92%+ detection accuracy.
- Applied machine learning techniques to analyze complex datasets and extract actionable insights and applied deep learning techniques to develop and train machine learning models for AI/ML projects.

## Core Skills

Python, SQL, Rust, TensorFlow, PyTorch, Keras, spaCy, BERT, YOLOv5, YOLOv8, Apache Spark, PySpark, Kafka, Airflow, Docker, Kubernetes, FastAPI, MLflow, nlp, Data Structures, Systems Design, test-driven development, mentor, large language models, Financial Services, Data Engineering, QA Testing, Snowflake, Calculus, pandas, numpy, scikit-learn, data science, linear algebra, mathematics, software engineering, software development, brainstorming, creative, Attention to Detail, constructive feedback, verbal communication skills, written and verbal communication, critical thinking, Problem solving, reliability

## Education

**Southeast Missouri State University**

Aug 2024 -

Masters in Applied Computer Science Computer Science (AI/ML)  
GPA: 3.83

## Certificates

**Microsoft Certified Solutions Expert**

**AWS Cloud Technical Essentials**

**Python Programming**  
*UMich*

**ML with Python**  
*IBM*

**Python for DS, AI & Dev**  
*IBM*